

Re-analysis of “Egress Thresholds and Wildfire Fatalities”

Fong et al., *PNAS* 2025 | doi:10.1073/pnas.2535081123 | replication review

Data: 38 fire × census-place records (SI sd01.xlsx). Numerator = place-attributed (“Identified”) fatalities; denominator = place population.

Issues found

- The headline figure regresses a cumulative sum against its own sort key.** Figure S1’s y-axis `cum_per_fatal` is a *reverse cumulative sum* of per-capita fatality rate, ordered by exits.
 - Monotonically decreasing by construction** (sum of non-negative increments): pinned at the grand total at fewest exits and **exactly 0** at most exits, so a negative slope is guaranteed regardless of the data — it appears even under a permuted (null) relationship.
 - A piecewise **“breakpoint”** is **nearly inevitable** for such a curve; the reported ~ 6.3 exits / 66.6th pct merely marks where the right-skewed exit distribution concentrates.
 - Points are **not independent** (each is a running total of the others), so the reported SEs, R^2 , and breakpoint CI are not interpretable.
- Weighting is unstable.** Across weighting schemes the per-capita slope slides $\sim 2.5\times$ (unweighted $R^2 \approx 0.09 \rightarrow$ population-weighted $R^2 \approx 0.19$ but slope *halves*). OLS-on-a-ratio is the wrong model for noisy counts.
- Six population denominators are wrong — systematically.** For destructive older fires the listed population is the *post-fire depopulated* census, not the at-risk population. **All six biased communities are low-exit (≤ 6) and all are inflated**, pushing toward the paper’s conclusion; two are the dataset’s #1/#2 highest rates.

Fire / place	exits	deaths	listed $\rightarrow$ corrected	rate inflation
Camp / Paradise	6	66	7,730 $\rightarrow$ 26,800	3.5 $\times$
Camp / Concow	6	9	306 $\rightarrow$ 710	2.3 $\times$
Carr / Keswick	3	4	141 $\rightarrow$ 451	3.2 $\times$
Butte / Mountain Ranch	2	2	256 $\rightarrow$ 1,628	6.4 $\times$
Mtn View / Walker	2	1	401 $\rightarrow$ 704	1.8 $\times$
Waldo Canyon / Manitou	3	2	235 $\rightarrow$ 4,309	$\geq 20\times$ (misassigned*)

*Manitou: the deaths occurred in the Mountain Shadows neighborhood of Colorado Springs, not Manitou Springs — a geographic misassignment, not just a stale vintage.

Impact on the authors’ interpretations

- “Threshold at ~ 6 exits” — not supported.** An artifact of the cumulative transform; a correct count model shows a *smooth, constant-per-exit* decline with no kink.
- Figure S4 per-capita significance — does not survive correction.** `fatality_pct` $\sim$ `exits` on corrected denominators moves from $p = 0.036$ to $p = 0.060$ (slope -28%). The paper reported $p = 0.048$.
- Causal / policy reading — unsupported.** Observational; ~ 35 communities; confounding (exits proxies small/remote/rural type; their Fig S3 `pop` $\sim$ `exits` $R^2 = 0.43$); selection on fatal events; fuzzy tiny-CDP denominators.

What does hold up (negative binomial + jackknife)

Counts modeled directly: `deaths` $\sim$ `exits` + `offset(log pop)`, negative binomial (Poisson is badly overdispersed, Pearson $\varphi \approx 16$, $\alpha \approx 1.1$).

	rate ratio / exit	effect	p
Original populations	0.813	-18.7% / exit	< 0.0001
Corrected populations	0.837	-16.3% / exit	< 0.0001

- Direction is real and survives everything** — the cumulative-artifact fix, the six denominator corrections, and overdispersion (quasi-Poisson $p = 0.0014$).
- Jackknife (leave-one-out) is rock-stable:** coefficient range -0.196 to -0.168 across all 38 deletions; the most influential point is *Altadena*, whose removal *strengthens* the effect. No single community drives the result.
- The corrected OLS per-capita slope is LOO-stable but leans on un-benchmarkable tiny CDPs (Moskowite Corner 109, Igo 148, Ono 72) — another reason to prefer the count model.

Verdict

“Fewer egress roads $\rightarrow$ higher per-capita wildfire fatality” is supported — modestly ($\sim 16\%$ per road), robustly, on corrected denominators. **The paper’s specific claims — a 6-exit threshold and the Figure S4 per-capita significance — are not.** No causal or policy conclusion is warranted.

Artifacts: idempotent corrections + corrected dataset in `data/fire-fatality-corrected.csv`. Census figures verified against U.S. Census 2010/2020 profiles and contemporaneous reporting.